

ONCOGENE ADVANCED DIAGNOSTICS LABORATORY

Multi-Cancer Early Detection (MCED) Next-Gen Liquid Biopsy Panel

CLIA ID: 24D2094819
CAP Accredited: #8491028
Clinical Protocol: MCED-v2.4

PATIENT IDENTIFICATION

Legal Name: Marcus Aurelius Vance
Date of Birth: 1958-09-12
Medical Record #: MRN-6192840
Gender / Sex: Male (Assigned at Birth)
Residential Address: 1482 Highland Parkway, St. Paul, MN 55116
Contact Phone: (651) 555-0194

SPECIMEN & ORDER INFORMATION

Accession ID: ACC-2026-MCED-8841
Collection Date: 2026-08-28 09:15 CDT
Received Date: 2026-08-29 07:45 CDT
Report Date: 2026-09-02 14:30 CDT
Ordering Physician: Dr. Sarah Lin, MD
Clinical Facility: Midwest Oncology Associates - Main Campus

OVERALL CANCER SIGNAL STATUS: **SIGNAL DETECTED**

Primary Predicted Signal Origin (CSO 1): **Pancreato-Biliary (Probability Score: 0.78)**
Secondary Predicted Origin (CSO 2): Upper Gastrointestinal (Probability Score: 0.16)
Estimated ctDNA Tumor Fraction (%): **5.2% (High Positive Circulating Load)**
Signal Confidence Classification: Tier 1 - High Specificity Genomic & Epigenetic Concordance

CLINICAL INTERPRETATION & ACTIONABLE SUMMARY

A circulating cell-free DNA (cfDNA) cancer signal was DETECTED in the plasma specimen. Machine learning pattern analysis of high-depth targeted somatic sequencing combined with whole-genome bisulfite methylation profiling indicates a primary tissue of origin localization of Pancreato-Biliary with a calibrated probability index of 0.78.

Multiple pathogenic somatic driver mutations in pancreato-biliary-associated lineage pathways were confirmed at high analytical confidence with an overall estimated circulating tumor fraction of 5.2%. Diagnostic workup including high-resolution targeted colonoscopy / cross-sectional imaging (CT Abdomen/Pelvis with IV contrast) is recommended.

Electronically Verified & Released By:
Dr. Nathan K. Sterling, MD, PhD, FCAP
Medical Director of Molecular Pathology | NPI: 1948201948

Diagnostic Facility:
OncoGene Diagnostics - Central Laboratory
9200 Medical Center Way, Suite 400, Minneapolis, MN

SECTION 1: TARGETED SOMATIC GENOMIC ALTERATIONS (ctDNA NGS)

Gene Symbol	DNA / Amino Acid Alteration	Target Exon	Variant Fraction (VAF %)	Clinical Significance / Pathogenicity Tier	Status
KRAS	c.35G>T (p.Gly12Val / G12V)	Exon 2	4.2%	Tier I - Pathogenic (Pancreato-Biliary / Upper GI)	DETECTED
SMAD4	c.1082G>A (p.Arg361His / R361H)	Exon 9	2.8%	Tier I - Pathogenic (TGF-beta Signaling Pathway)	DETECTED
TP53	c.524G>A (p.Arg175His / R175H)	Exon 5	3.5%	Tier I - Pathogenic (Structural Hotspot Mutation)	DETECTED

Non-Detected Genes: BRAF, NRAS, PIK3CA, EGFR, ERBB2, MET, RET, ALK, ROS1 (Below analytical sensitivity limit of 0.1% VAF)

SECTION 2: EPIGENETIC METHYLATION PATTERNS & CIRCULATING PROTEIN MARKERS

Biomarker / Assay Target	Biological Modality	Quantitative Index / Value	Reference Cutoff Baseline	Result Interpretation
SEPT9 Promoter Methylation (sept9_index)	Targeted Bisulfite	0.82	< 0.20 (Negative Baseline)	ELEVATED HYPERMETHYLATION
SDC2 Promoter Methylation (sdc2_index)	Targeted Bisulfite	0.74	< 0.25 (Negative Baseline)	ELEVATED HYPERMETHYLATION
NDRG4 Promoter Methylation (ndrg4_index)	Targeted Bisulfite	0.65	< 0.30 (Negative Baseline)	ELEVATED HYPERMETHYLATION
Serum Carcinoembryonic Antigen (CEA)	Immunoassay ECLIA	18.9 ng/mL	< 3.0 ng/mL (Non-Smoker)	ELEVATED CIRCULATING

ASSAY METHODOLOGY, ANALYTICAL SENSITIVITY & REGULATORY DISCLAIMERS

Methodology: High-throughput sequencing of circulating cell-free DNA (cfDNA) isolated from peripheral whole blood EDTA tubes.
Analytical Performance: Somatic SNV/Indel limit of detection (LOD) = 0.1% VAF at 50,000x raw depth. Methylation analytical specificity = 99.2%.
Regulatory Notice: Developed and validated pursuant to CLIA regulations. This test is intended to be used in conjunction with clinical diagnostic guidelines.